Using Large Language Models to identify Narcissism based on texts

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Notes：Zhuoran Tu and Zhaohui Zhang denote equal contribution to the work, are listed as co-first author

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Abstract

Open-ended narcissistic assessments can reduce the distortion issues in evaluations caused by factors such as social desirability and self-deception. However, due to limitations in scoring costs and efficiency, they are seldom used in research and actual selection processes. Large language models with high-level text processing capabilities offer a new method for the automated scoring of test texts, enabling more accurate assessments without relying on large-scale labeled datasets. However, no studies have yet explored their accuracy in narcissism prediction. This study analyzes the effectiveness of Large Language Models (LLMs) in open-ended narcissism assessment by examining the response texts from 1,669 participants in the open-ended narcissism situational judgment test and the open-ended narcissism conditional reasoning test. We used human expert scoring as the gold standard and analyzed the scoring effectiveness of the ChatGPT model from multiple perspectives including sentence-level/answer level and zero-shot/few-shot. We also compared it with machine learning models. In addition, we systematically compared the scoring characteristics of human experts with those of ChatGPT and found that ChatGPT possesses rater-like characteristics. In sentences where human raters are prone to scoring errors, ChatGPT is also likely to make mistakes. The difference in correct prediction rates was significant, with the effect size measured by Cohen's d coefficient ranging between 0.56 and 1.12. Furthermore, in the detailed error analysis, we also found that ChatGPT exhibited a similar pattern of scoring errors to human experts. Overall, our study demonstrates the potential of LLMs in identifying narcissistic personalities in limited datasets.

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1. *Introduction*

The generality of large language models (LLMs) enables them to perform text scoring tasks across multiple scenarios without requiring additional training (Rathje et al., 2023). These models also facilitate the identification of personality traits based on texts (Zhang et al., 2024). Open-ended psychological tests require participants to describe their thoughts and ideas in textual form, providing more information related to assessed traits compared to self-report inventories (Connor & Reimers, 2019). However, these tests are costly and are rarely used in research and applications. While machine learning-based short-text automatic scoring techniques can effectively reduce the cost of open-ended questions (Cınar et al., 2020), achieving satisfactory measurement accuracy remains challenging due to the paucity of large-scale labeled databases (Jakel et al., 2008). Large language models present a novel approach to mitigating the cost problem associated with open-ended question scoring. These models evaluate responses based on trained knowledge, free from constraints imposed by databases or assessment scenarios. Furthermore, they exhibit remarkable precision and generalization capabilities in zero-shot learning contexts (Lee et al., 2024). Researchers have explored the effectiveness of using large language models with non-test text sources such as internet content and likes data on Facebook alongside test/interview text for the assessment of personality traits. (Rathje et al., 2023; Zhang et al., 2024). The results show that LLMs achieve accuracy comparable to machine learning models, underscoring their significant potential for precise personality assessment.

Narcissism is a well-known psychological construct that is both an independent personality disorder included in the diagnostic criteria of DSM-5 (APA, 2013) and one of the "dark triad" of personality traits, along with Machiavellianism and psychopathy (Paulhus & Williams, 2002). Furthermore, it is widely recognized as one of the aversive personalities in interpersonal relationships (Kowalski, 2003). Given its profoundly destructive impact on both organizations and individuals (Carpenter, 2012; Kauten & Barry, 2014; Miller et al., 2010; Miller et al., 2017), assessing narcissism is crucial (Emmons, 1987; Krizan & Herlache, 2018; Raskin, 1978). However, narcissism itself carries implications for social evaluation, making self-report assessments susceptible to distortions such as self-deception (Sackeïm, 1979) and social desirability response bias (van de Mortel, 2008). These distortions can potentially compromise the accuracy of measurements in narcissism research.. Situation judgment tests (SJT) and conditional reasoning tests (CRT) elicit participants' natural behavioral responses that align with assessment

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objectives, thereby mitigating social desirability bias and facilitating covert assessment (Webster et al., 2020; LeBreton et al., 2020). Traditional situation judgment tests, which is based on multiple-choice questions, may restrict the diverse expressions of participants. This format lead participants to focus more on selecting options rather than understanding the questions, thus lacking predictive validity for real behaviors (Kormos & Gifford, 2014; Rockstuhl et al., 2015). Open-ended formats allow participants to express themselves freely but are hindered by scoring costs. Leveraging automated scoring methods like LLMs holds promise in effectively scoring open-ended scenario judgment and conditional reasoning tests. The implementation of such methods could potentially overcome the limitations of traditional scoring techniques while maintaining the benefits of open-ended assessments.

This study aims to explore the effectiveness of LLMs in assessing narcissistic personality traits, investigating whether these models can accurately predict narcissistic personality traits in SJT and CRT without prior training. The structure of the article is as follows. In Section 2, we will introduce the breakthrough of LLMs in traditional text automatic scoring techniques, current research on LLMs automatic scoring, limitations of current narcissism assessment methods, feasibility of open-ended assessments, and based on existing research, propose specific research questions. Section 3 provides a detailed description of data and methods used in our study. Section 4 presents the results of our study, followed by a thorough discussion of the findings in Section 5.

2. Literature review

2.1 LLMs for Personality Prediction Based on Textual Materials

Personality prediction based on textual materials represents a prominent research area intersecting psychology and computer science. Prior to the advent of LLMs, numerous machine learning models for personality and psychological state prediction had been successfully developed and implemented. These models encompass various personality traits and psychological issues, including depression (Liu et al., 2022), anxiety (Richter et al., 2021), PTSD (Ramos-Lima et al., 2020), the Big Five personality traits (Xue et al., 2018), and emotions (Kratzwald et al., 2018). These data-driven models for personality prediction exhibit strong performance but suffer from several limitations. Firstly, they are heavily reliant on the datasets used (Jäkel et al., 2008). Establishing machine learning models for personality prediction requires large-scale, labeled datasets, and models trained on specific datasets often struggle to generalize well to other datasets. Additionally, existing textual datasets predominantly originate from internet texts

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or social media likes (Pulicharla, 2024). Despite their vast volume, these datasets often contain limited information directly relevant to personality traits, thereby resulting in lower accuracy in assessments. In addition to these models, researchers have also explored methods based on short texts for identifying personality traits or mental health indicators. For instance, He et al. (2017) developed an automated scoring model for PTSD based on open-ended questions. Compared to big data approaches, short text analysis potentially captures richer information, aiming for more precise assessments. However, due to the smaller dataset sizes, training effective machine learning scoring models that generalize well remains challenging, often leading to issues of overfitting and underfitting. As a result, some researchers have employed lexicon-based approaches for personality prediction. For example, Chung & Pennebaker (2018) utilized the LIWC (Linguistic Inquiry and Word Count) dictionary to assess personality traits based on textual content. Nonetheless, these methods still face limitations in achieving high accuracy.

LLMs trained with parameter counts in the billions exhibit the phenomenon of "emergent intelligence," demonstrating the ability to perform well across multiple diverse tasks without requiring additional training (Rathje et al., 2023). This capability holds promise for addressing existing challenges in text-based automated recognition models. Researchers have endeavored to apply LLMs to the identification of mental health issues using internet-based texts. For instance, Xu et al. (2024) attempted to assess users' mental health using LLMs with online web text data. They evaluated the performance of five LLMs, including Alpaca, Alpaca-LoRA, FLAN-T5, GPT-3.5, and GPT-4, under zero-shot, few-shot, and instruct finetuning conditions across five different sources of web text datasets. The results indicated that LLMs demonstrate high scoring accuracy, particularly following instruct finetuning. Rathje et al. (2023) utilized the GPT model to analyze textual materials across 12 different language backgrounds for levels of psychological traits (sentiment, discrete emotions, offensiveness, and moral foundations). Their study findings also indicated that the GPT model exhibits strong performance, achieving or approaching the levels of top-performing machine learning models in specific tasks. In addition, LLMs have also demonstrated good performance in the automatic scoring of short texts (citattion). Lee et al. (2024) addressed the grading of student responses in scientific education by designing a questioning method involving prompt writing, reviewing, validating, revising, and testing (WRVRT). They systematically investigated the impact of zero-shot learning (without any examples in the prompt) and few-shot learning (with a few examples in the prompt) on scoring accuracy. The results indicated that with appropriate

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prompt design, LLMs can effectively handle the grading of student responses in scientific education. Zhang et al. (2024) utilized LLMs to assess users' HEXACO personality traits based on structured interview texts. They found that LLMs, without any specific training, demonstrated validity performance comparable to or even better than task-specific AI models. These series of studies collectively highlight the significant potential of LLMs in predicting psychological traits based on textual data, encompassing both online content and test short-answer texts. A comprehensive review article published in Nature Review Psychology posits that LLMs will provide new avenues for measurement in multiple domains of psychology such as affective, cognitive, developmental, and metascience, as well as interdisciplinary research involving psychology (Demszky et al., 2023).

2.2 Narcissism and Assessment

As defined by Pincus (2008), narcissism refers to the individual's capacity to maintain a positive self-image during processes of self-regulation, emotional regulation, and interpersonal interactions. The impact of narcissism can be multifaceted (Wink, 1991; Morf & Rhodewalt, 2003). On one hand, narcissism is often associated with behaviors such as exploitation of power and control over others (Raskin, 1978; Emmons, 1987), leading to severe negative interpersonal consequences (Miller et al., 2017). This personality trait is closely linked to aggression (Kauten & Barry, 2014), antisocial behavior (Carpenter, 2012), and borderline personality disorder (Miller et al., 2010). On the other hand, individuals with narcissism demonstrate heightened vulnerability when confronted with information threatening their self-image, a phenomenon referred to as narcissistic injury (Di Sarno, 2020). Compared to non-narcissistic individuals, those with narcissistic traits are more prone to experiencing tendencies towards depression, feelings of emptiness, and suicidal tendencies (Miller & Campbell, 2008; Tritt et al., 2011; Zerach, 2020).

Currently, assessments of narcissism in psychology primarily rely on self-report scales (Krizan & Herlache, 2018). The Narcissistic Personality Inventory (NPI) and Pathological Narcissism Inventory (PNI) are widely used tools in both research and clinical diagnosis of narcissistic personality (Miller et al., 2017; Pincus et al., 2009). Self-report scales offer a convenient and efficient method for assessing narcissism. However, they have been subject to considerable debate due to concerns regarding systematic biases and issues pertaining to their structural validity, as consistently highlighted by researchers and reviewers alike. The responses to self-report scales can be influenced by various factors, including language comprehension, information processing

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capabilities, and motivational mechanisms. (Chan, 2010)， Especially when assessments involve negative evaluations of oneself, respondents are more susceptible to measurement distortions due to self-deception (Sackeïm, 1979) and social desirability response bias (van de Mortel, 2008); Furthermore, self-report scales lack prediction validity. A meta-analysis examining the relationship between self-report scales and objective behavior indicated that only 21% of the variance in self-report measures can be attributed to objective behavior (Kormos & Gifford, 2014).

To achieve precise assessment of narcissism, it is essential to minimize the influence of self-deception and social desirability response bias. SJT and CRT, commonly used in organizational and industrial psychology, are effective measurement techniques designed to mitigate the risk of false responding. SJT is a type of assessment based on real-life scenarios. Its fundamental format involves presenting participants with specific situational contexts wherein they are asked to respond based on their prior experiences or personal behavioral tendencies. This test aims to measure objectives in a more covert manner, capturing nuanced aspects of behavior and decision-making within defined contexts. SJT are typically designed in a multiple-choice format, which to some extent limits the freedom of expression for participants. Rockstuhl et al. (2015) noted that SJTs primarily focus on evaluating the effectiveness of various response options rather than exploring how individuals perceive and interpret situations. Rockstuhl et al. (2015) employed verbal protocol analysis, collecting spoken reports from participants to understand their interpretations of situational contexts. They coded and analyzed language content from the perspectives of 11 situational factors, including intentions, emotions, and thoughts. In their longitudinal empirical study, Rockstuhl and colleagues demonstrated that allowing participants to respond freely to situations effectively predicted future job performance.

The CRT is a standardized implicit projection technique based on the premise that individuals tend to reason and attribute according to their own cognitive styles, also known as defensive mechanisms. By observing participants' reasoning in a fictitious logical reasoning test, CRT assesses their underlying personality traits (Baxley et al., 2024; LeBreton et al., 2020). CRT serves as a more covert method for personality assessment, effectively circumventing the influences of self-deception and social desirability bias. However, similar to SJT tests, the multiple-choice format of CRT somewhat limits the freedom of expression for participants. According to Connor Desai and Reimers (2019), participants in open-ended Narcissism SJT and CRT tests are expected to provide richer information relevant to narcissism. The

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utilize of LLMs for narcissistic personality recognition in these test texts holds promise for developing a narcissism assessment tool that is not only efficient and accurate but also suitable for large-scale assessment scenarios.

2.3 Research Question

Assessing narcissism, which involves negative self-evaluation, is prone to distortion from self-deception and social desirability biases. Overcoming these systematic biases to achieve accurate narcissism assessment continues to pose a significant challenge within the field of psychology. Utilizing machine learning methods for automatic scoring of short test texts presents a promising approach to address these issues. However, the lack of large-scale labeled datasets limits its practical application. LLMs offer a solution by providing a low-cost, text-based automatic scoring method that does not depend on extensive labeled datasets or require training. Several personality assessment tasks leveraging LLMs have demonstrated the feasibility of using LLMs for personality prediction (Rathje et al., 2023).

We aim to apply LLMs to the task of narcissistic personality recognition based on psychological test texts (SJT and CRT), comparing its performance with top-performance machine learning models tailored for specific tasks. We seek to investigate the impact of zero-shot learning and few-shot learning on scoring accuracy. Additionally, we aim to compare LLMs' scoring results with expert evaluations for a more detailed analysis, exploring similarities and differences between LLMs and human expert ratings. This study intends to provide guidance and recommendations for the use of scoring models. Specifically, our research questions include:

RQ1： Can LLMs accurately predict the narcissistic personality based on open-ended SJT and CRT response texts, and how can the accuracy of LLMs in open-ended psychological tests be effectively improved?

RQ2： How does the scoring pattern of large language models compare to that of human experts, and what are the consistent and differences between LLMs and human expert?

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3. Method

3.1 ***Measurement***

3.1.1 Open-ended Narcissism Situation Judgement Test

Tu (2021), based on Ronningstam's (2020) research, developed a situational judgment test targeting narcissism with four underlying processes: Limited Affect Tolerance, Perfectionism, Competitive Sense of Control, and Conditional Self-Esteem. This test includes 16 multiple-choice questions and 4 open-ended questions, demonstrating good reliability and validity among both university students and corporate employees. The Cronbach's alpha reliability for the four dimensions ranges from 0.59 to 0.89, and they exhibit moderate to significant correlations with other mental health constructs, with Pearson correlation coefficients ranging from 0.18 to 0.37. We will utilize the open-ended question measuring Limited Affect Tolerance from this test, and conduct narcissism scoring within Tu's (2021) theoretical framework. The question is as follows: "You are a management consultant who frequently provides various management consulting services to companies. During one of your services, based on your professional expertise and experience, you provided detailed management advice to a company, which then implemented adjustments according to your recommendations. However, after the company implemented these management changes, the situation worsened rather than improved. Consequently, the company hired another management consultant from your firm, whose advice led to significant improvements in the company's management. Subsequently, you discover that your colleagues within the company privately question your competence because of this incident. In this situation, what are your emotional responses and how would you respond?"

3.1.2 Open-ended Narcissism Conditional Reasoning Test

Feng (2019) developed the CRT-N-15, a reliable and valid assessment tool, and validated it within a Chinese university student population. The internal consistency coefficient for the 15 items is reported as 0.43, which is comparable to the reliability metrics of many other CRT tests, often ranging from .40s to .50s (LeBreton et al., 2020). The discriminability indices exhibited a range from 0.28 to 0.66. We have adapted the highest-discriminating question from Feng (2019)'s CRT-N-15 into an open-ended format, and based on Feng's theoretical framework, expert evaluators will assess the responses. The question used is as follows: "How would you argue against the viewpoint

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that 'elites are always resented because they make ordinary people feel inferior'?"

3.2 Participants

This study collected responses from participants across three batches, totaling data from 1651 participants. The first two batches involved large-scale data collection, serving as the training data for the study. The third batch, a smaller-scale collection, was used as the testing data for the research.

In the first batch, we collected responses from 775 participants in the Narcissism Conditional Reasoning Test (CRT). These participants were primarily sourced from a corporate selection process. The average age of these participants was 23.19 years, and the female proportion was 56.5%.

In the second batch, we collected responses from 794 participants in the Narcissism Situational Judgment Test (SJT). These participants were primarily recruited through the internet. The average age of these participants was 31.39 years, and the female proportion was 50.5%.

In the third batch, we recruited 100 participants who completed both the Conditional Reasoning Test and the Situational Judgment Test. This batch served as the testing set to evaluate the performance of all models in predicting narcissism. After removing 6 invalid responses (unmeaningness/irrelevant texts or answered by chatbot), 80.9% of the participants were aged between 18 and 25 years, and the female proportion was 53.2%.

3.3 Rules of Text Coding and Scoring

Starting from the narcissism theoretical frameworks proposed by Tu (2021) and Feng (2019), responses to specific open-ended questions are scored based on different levels of narcissism expression. The overall coding rules are structured as follows: firstly, sentences in the response text are segmented into distinct semantic units. Next, experts classify these sentences according to a coding scheme. Finally, the response text is evaluated based on the distribution of sentence categories within each response. In the open-ended situation judgment test responses, a total of 10 sentence categories reflecting different levels of narcissism were identified (detailed information can be found in Table A1 in the Appendix). Sentence categories scoring 0 points include "Analyze the reasons for failure," "Accepting failure calmly," and "Adaptive emotional adjustment." Sentence categories scoring 1 point include "Seeking social support" and "Compensation." Sentence categories scoring 2 points include "Dismiss," "Avoidance," and "Shame blaming." In the Open-

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ended Conditional Reasoning Test, 12 distinct sentence coding categories were extracted to reflect varying degrees of narcissistic bias (detailed information can be found in Table A2 in the Appendix).

Subsequently, scoring in the test will assign scores based on the presence of sentences indicating levels of narcissism in each response. In the SJT subjective questions, scoring is based on the highest-rated sentence reflecting narcissism in the respondent's text. In CRT subjective questions, responses are scored based on the number of distinct narcissistic thinking patterns present. A score of 0 is assigned when no narcissistic cognitive biases are evident, 1 point is awarded when at least one typical narcissistic cognitive bias is present, and 2 points are allocated when two or more distinct narcissistic cognitive biases are identified.

3.4 Text Scoring Prediction based on Machine Learning Methods

Two distinct prediction strategies are employed. The first strategy predicts categories (labels) for each sentence using a machine learning model. Scores for narcissism in SJT and CRT response texts are derived using the same composite scoring method as human expert ratings (**Sentence-Level Model**). The second strategy directly predicts narcissism scores using a machine learning model based on entire response texts (**Answer-Level Model**).

Two sentence features are utilized as features for machine learning modeling. The TFIDF and BERT word embeddings are employed as textual features: the former calculates the importance of words based on their frequency in the text, capturing lexical but not semantic features of sentences; the latter, BERT embeddings, leverage a pre-trained language model released by Google, incorporating semantic features of sentences. Specifically, we use the BERT-Chinese-base pre-trained model, extracting features from the weights of the second-to-last layer of the BERT neural network (comprising 768 dimensions in Bert-Chinese-base). This approach involves utilizing these embeddings directly without fine-tuning the BERT model for subsequent modeling tasks.

Four machine learning methods were utilized for constructing the scoring models: logistic regression, support vector machine, random forest, and fully connected neural network. To fit the best-performing models, each scoring model will undergo training where the training dataset is split into training and validation sets. A 5-fold cross-validation method will be employed to search for optimal hyperparameters of the machine learning models within the training and validation sets. Subsequently, based on these optimal hyperparameters, automatic scoring models will be

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established using all training data. The top-performing machine learning models, combining the best-performing models from the two different text features (TFIDF and BERT embeddings), will be selected for comparison with LLMs.

3.5 Text Scoring Prediction based on ChatGPT

Based on ChatGPT 3.5-0301 and ChatGPT 4-0613, two LLMs, we aim to develop a text-based narcissism scoring model. The overall scoring process will involve setting the temperature parameter to 1. Two distinct scoring methodologies will be employed during data collection: the first method mirrors expert evaluation processes by initially encoding and categorizing sentences, synthesizing scores based on sentence evaluations (**Sentence-Level**); the second method directly assigns scores to entire response texts (**Answer-Level**). Prompt design will incorporate both Zero-shot and Few-shot methods simultaneously.

Zero-shot approach involves providing ChatGPT with task descriptions and sentence categories for classification during the scoring process, without offering any specific examples. ChatGPT directly classifies sentences based solely on the provided task description and categories.

Few-shot approach involves initially providing ChatGPT with typical examples of different sentence categories before conducting the classification task. Unlike traditional methods where examples are presented all at once, we adopt a multi-turn dialogue approach. In this approach, ChatGPT is presented with example sentences gradually through rounds of interaction. ChatGPT classifies each example sentence and receives feedback regarding the category to which each sentence belongs. At least one typical example is provided for each sentence category.

3.6 Comparative analysis between LLMs and human expert

In comparing human expert and large-scale model (LLMs) scoring at the sentence level, an analysis is conducted focusing on differences and similarities across two dimensions: sentence confusion and error patterns within sentence categories.

Sentence Confusion Analysis: Sentences were classified to 2 different groups based on the initial classification agreement between two experts. Sentences are categorized into high confusion group when initial disagreement between experts and low confusion group when initial agreement between experts. The accuracy differences of LLMs between different confusion levels are examined.

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Error Pattern Analysis: Error patterns are categorized into two distinct types, E1 and E2. E1 denotes classification errors within a specific category (i.e. predict the sentences which not belong to A1 to A1), while E2 refers to misclassifying a sentence into a different category (i.e. predict the sentence which belong to A1 to other labels). By comparing the proportions of E1 and E2 errors between human experts and LLMs across various categories, the differences and similarities in error patterns are explored.

This analysis provides insights into the performance discrepancies between LLMs and human experts in sentence-level narcissism scoring, shedding light on their respective strengths and weaknesses in handling sentence confusion and error patterns within different sentence categories.

4. Result

4.1 Expert Rating Results

Six second-year psychology graduate students participated in the coding and rating of texts: 2 experts handled coding for the SJT texts, while 4 experts managed coding for the CRT texts. In the SJT texts, inter-rater reliability among the two experts in the Training data was 0.90, and in the Testing data, it was 0.83. For the CRT texts, inter-rater reliability among the four experts in the Training data was 0.65, increasing to 0.85 in the Testing data. Initially, due to the lack of a unified coding framework, there was lower agreement among experts in CRT text coding and scoring. However, after establishing a standardized framework, the coding consistency in the Testing data reached acceptable levels. All texts with inconsistent coding were re-evaluated by experts in focus groups to ensure accuracy and consistency. Based on expert coding results, participants' narcissism scores in the SJT and CRT tests were obtained.

Table 1 presents the scoring distribution for SJT and CRT response texts.

Table 1 scores of answers in SJT open-ended questions

	SJT		CRT	
Score	Training data	Test data	Training data	Test data
-1(Invalid answer)	26	1	15	5
0	533	65	363	37
1	138	19	327	41
2	97	9	70	11
Total	794	94	775	94

4.2 Sentence-Level Model based on ML and LLMs

4.2.1 Sentence-Level Machine Learning Models

Table 2 presents the average Precision and Recall of four different machine learning models

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(LR, SVM, RF, NN) across three prediction categories. As indicated in the table, in the SJT task, all four models exhibit higher Precision and lower Recall. Specifically, the LR model utilizing TFIDF features performs the best among them. In contrast, for the CRT task, Precision and Recall are relatively balanced across models. Again, the LR model using TFIDF features shows relatively superior performance compared to others.

These findings suggest that in the SJT task, the models tend to prioritize Precision over Recall, indicating conservative predictions. In the CRT task, the models achieve a more balanced Precision-Recall trade-off. The LR model's use of TFIDF features consistently demonstrates optimal performance across both tasks according to the metrics provided in Table 2.

Table 2 Overall Performance Table of Sentence-Level Machine Learning Scoring Models

Model	SJT				CRT			
	TFIDF		BERT		TFIDF		BERT	
	Precision	Recall	Precision	Recall	Precision	Recall	Precision	Recall
LR	**0.91**	**0.57**	0.82	0.59	**0.70**	**0.71**	0.63	0.37
SVM	0.93	0.49	0.77	0.58	0.73	0.66	0.65	0.61
RF	0.82	0.45	0.24	0.32	0.68	0.63	0.62	0.60
NN	0.86	0.53	0.81	0.60	0.65	0.60	0.54	0.46

We continue to compare the performance of these two machine learning models in specific scoring aspects, as shown in Table 3. In the SJT task, the models demonstrate high Precision and Recall in the 0-point category, at 0.82 and 0.95 respectively. However, for the 1-point and 2-point categories, they exhibit notably high Precision but low Recall, indicating a cautious approach towards diagnosing high narcissism scores and a significant rate of false negatives. In the CRT task, the models exhibit relatively balanced Precision and Recall across the 0-point, 1-point, and 2-point categories, although they do not reach a highly accurate level.

Table 3 Top-performance Sentence-Level Machine Level Models Metrics

Score	**SJT**		**CRT**	
	Precision	**Recall**	**Precision**	**Recall**
0	0.82	0.95	0.65	0.71
1	0.91	0.53	0.69	0.61
2	1.00	0.22	0.75	0.82

4.2.2 Sentence-Level LLMs

Table 4 presents the performance evaluation of the GPT-3.5-turbo-0301 and GPT-4-0613 models on SJT and CRT sentence categories. Overall, the GPT-4-0613 model trained via Few-shot exhibits superior performance compared to other models, achieving the highest Precision and Recall rates, surpassing the top-performance Machine Learning Models as well.

Table 4 Overall Performance Table of Sentence-Level LLMs

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Model	SJT				CRT			
	Zero-shot		Few-shot		Zero-shot		Few-shot	
	Precision	Recall	Precision	Recall	Precision	Recall	Precision	Recall
GPT350301	0.73	0.73	0.70	0.72	0.75	0.54	0.66	0.54
GPT040613	0.77	0.71	**0.88**	**0.75**	0.57	0.50	**0.76**	**0.60**

Based on specific performance details, the results are shown in Table 5. In the SJT task, the GPT-4-0613 model trained via Few-shot Learning significantly outperforms the machine learning models. Although there are still some missed detections (false negatives) in the 1-point and 2-point categories, the overall proportion of missed detections is lower, with Recall rates of 0.63 and 0.67 respectively, and the Precision rates do not decrease significantly. In the CRT task, similarly, the GPT-4-0613 model version trained via Few-shot Learning shows higher performance. Although there are missed detections for the 2-point category, overall performance is comparable to or better than that of the machine learning models.

Table 5 LLMs Sentence-Level Models Metrics

Models	Score	SJT				CRT			
		Zero-shot		Few-shot		Zero-shot		Few-shot	
		Precision	Recall	Precision	Recall	Precision	Recall	Precision	Recall
GPT350301	0	0.86	0.88	0.85	0.85	0.59	0.89	0.69	0.59
	1	0.83	0.53	0.56	0.53	0.67	0.54	0.62	0.68
	2	0.50	0.78	0.70	0.78	1.00	0.18	0.67	0.36
GPT040613	0	0.83	0.88	**0.85**	**0.94**	0.62	0.68	**0.78**	**0.76**
	1	0.73	0.58	**0.80**	**0.63**	0.60	0.63	**0.76**	**0.76**
	2	0.75	0.67	**1.00**	**0.67**	0.50	0.18	**0.75**	**0.27**

4.2.3 Performance of Sentence-Level Models with Varying Sample Sizes

To emphasize the advantage of LLMs models not relying on additional training samples, we plotted the changes in prediction accuracy, recall, and F1 scores with varying sample sizes. Figure 1 illustrates the variation in performance on the SJT task with respect to sample size. Overall, the metrics of the ML model gradually increase with increasing sample size, stabilizing and eventually surpassing the performance of LLMs.

For the SJT task:

1) Predicting 0 scores requires minimal training samples, approximately 500 training instances achieve performance comparable to the GPT040613 version model and stabilize thereafter.
2) Predicting 1 score shows a noticeable fluctuating upward trend, indicating potential for

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improvement even after training with all 1970 training sentences.

3) Predicting 2 scores displays instability even when all training sentences are used, significantly underperforming compared to LLMs.

These findings underscore the robustness and efficiency of LLMs in tasks where minimal training data is available, demonstrating their capability to achieve competitive performance without the need for extensive additional training samples.



Figure 1 Sentence-Level ML Model Metrics Change along with the sample of training data in SJT question (Green dash line is GPT040613 with few-shot, red dash line is GPT040613 with zero-shot)

Figure 2 illustrates the variation in metrics (accuracy, recall, and F1 score) for the CRT task with respect to sample size. Similar to the SJT task, overall trends show that metrics for ML models gradually increase with larger sample sizes and stabilize over time. Notably, around a sample size of 600, ML models achieve performance comparable to the GPT model. This could be attributed to the more balanced distribution of samples across the three categories in the CRT task.

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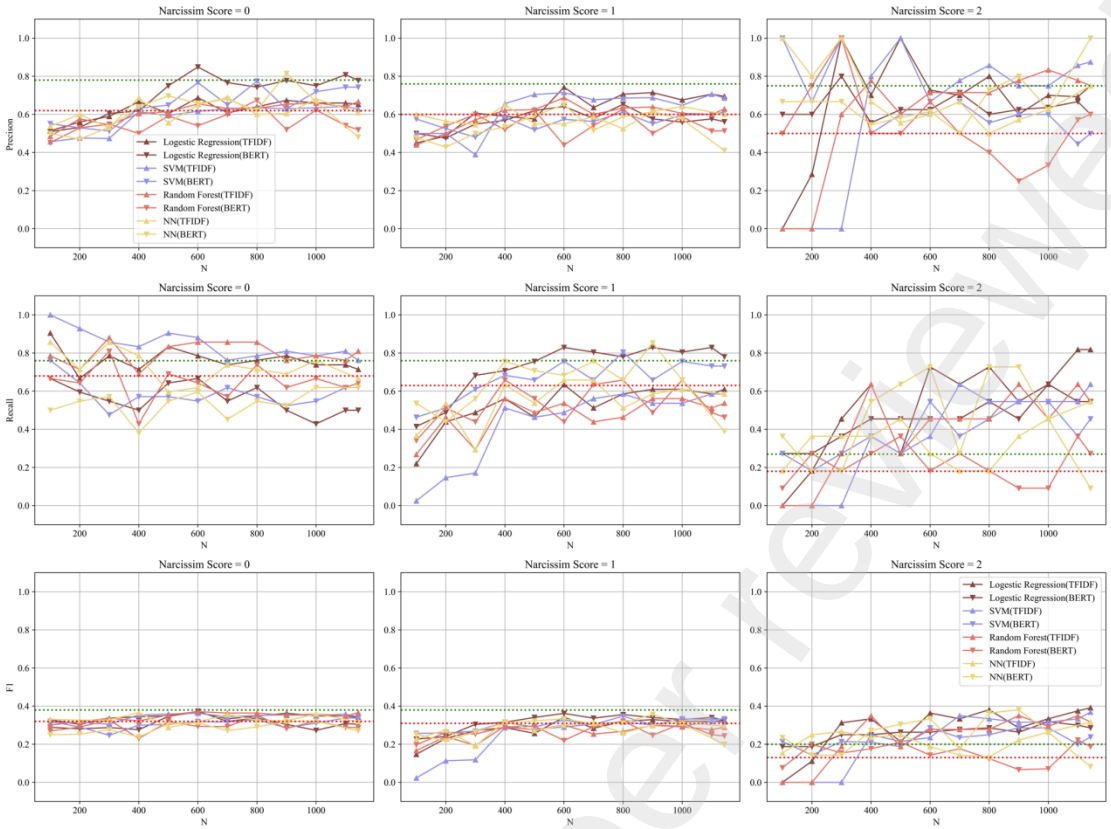


Figure 2 Sentence-Level ML Model Metrics Change along with the sample of training data in CRT question (Green dash line is GPT040613 with few-shot, red dash line is GPT040613 with zero-shot)

4.3 Answer-Level Model based on ML and LLMs

4.3.1 Answer-level Machine Learning Model

Table 6 presents the modeling results of machine learning models at the Answer-level. Numerically, the accuracy and recall of models at the Answer-level are notably lower compared to the Sentence-Level. This observation is understandable since Answers typically contain richer semantic content compared to Sentences. Across both tasks, LR models utilizing BERT embeddings achieved the best performance.

Table 6 Overall Performance Table of Answer-Level Machine Learning Scoring Models

Model	SJT				CRT			
	TFIDF		BERT		TFIDF		BERT	
	Precision	Recall	Precision	Recall	Precision	Recall	Precision	Recall
LR	0.57	0.52	**0.72**	**0.59**	0.39	0.46	**0.39**	**0.47**
SVM	0.58	0.44	0.62	0.51	0.38	0.46	0.32	0.38
RF	0.55	0.47	0.27	0.32	0.33	0.39	0.31	0.37
NN	0.46	0.42	0.90	0.39	0.37	0.44	0.30	0.36

Following this, we proceeded to analyze the performance of the LR model utilizing BERT embeddings across specific scoring levels. In the SJT task, the model exhibited higher accuracy and recall in predicting scores of 0, whereas for scores of 1 and 2, both accuracy and recall were notably

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lower. In the CRT task, the ML model completely struggled to evaluate scores of 2, and significant classification errors were observed in distinguishing between scores of 0 and 1.

Table 7 Top-performance Answer-Level Machine Level Models Metrics

Score	SJT		CRT	
	Precision	**Recall**	**Precision**	**Recall**
0	0.81	0.95	0.57	0.78
1	0.70	0.37	0.60	0.63
2	0.67	0.44	0.00	0.00

4.3.2 Answer-level LLMs

Table 8 presents the performance of GPT-3.5 and GPT-4 models at the Answer-Level under Zero-shot and Few-shot conditions. In contrast to the scoring at the Sentence-Level, we observed that GPT-040613 performs relatively better with Zero-shot Learning at the Answer-Level. Interestingly, providing examples seems to degrade performance in scoring tasks at this level.。

Table 8 Overall Performance Table of Answer-Level LLMs

Model	SJT				CRT			
	Zero-shot		Few-shot		Zero-shot		Few-shot	
	Precision	Recall	Precision	Recall	Precision	Recall	Precision	Recall
GPT350301	0.73	0.54	0.44	0.47	0.41	0.46	0.40	0.46
GPT040613	**0.80**	**0.56**	0.59	0.72	**0.66**	**0.45**	0.58	0.52

Table 9 displays the Confusion Matrix for Answer-Level LLMs. From the table, it is evident that LLMs perform poorly in sentences rated 1 and 2, with imbalanced Precision and Recall metrics indicating widespread issues of false positives and false negatives. Additionally, comparing Zero-shot and Few-shot scenarios reveals that Few-shot does not significantly improve prediction performance in SJT and CRT tasks. In fact, it reduces prediction efficiency in certain categories for LLMs.

Table 9 Answer-Level LLMs Metrics

Models	Score	SJT				CRT			
		Zero-shot		Few-shot		Zero-shot		Few-shot	
		Precision	Recall	Precision	Recall	Precision	Recall	Precision	Recall
GPT350301	0	0.88	0.46	0.86	0.28	0.58	0.86	0.60	0.70
	1	0.31	0.84	0.26	0.58	0.64	0.51	0.60	0.68
	2	1.00	0.33	0.19	0.56	0.00	0.00	0.00	0.00
GPT040613	0	**0.88**	**0.78**	1.00	0.52	**0.76**	**0.43**	0.79	0.73
	1	**0.52**	**0.79**	0.44	0.74	**0.60**	**0.44**	0.62	0.73
	2	**1.00**	**0.11**	0.32	0.89	**0.61**	**0.48**	0.33	0.09

4.3.3 Performance of Answer-Level Models with Varying Sample Sizes

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In the SJT task, the precision, recall, and F1 scores for scores of 1 and 2 fluctuate and increase with increasing sample sizes. Meanwhile, for the 0-score level, precision gradually increases with sample size, while recall decreases with increasing samples. This pattern is reasonable, indicating that with fewer samples, ML models only predict 0 scores but gradually start predicting scores of 1 and 2 as sample size increases. Comparing ML models with LLMs, both exhibit unstable and poorer performance in predicting scores of 1 and 2 under the condition of 800 samples.



Figure 3 Answer -Level ML Model Metrics Change along with the sample of training data in SJT question (Green dash line is GPT040613 with few-shot, Red dash line is GPT040613 with zero-shot)

In the CRT task, metrics exhibit a similar pattern of change as observed in the SJT task. For predictions of 0 scores, accuracy increases gradually with increasing sample size, while recall decreases with more samples. For predictions of 1 and 2 scores, both accuracy and recall improve steadily with increasing sample size. Specifically, in predicting 1 score, as the sample size approaches 800, the performance gradually approaches that of LLMs. However, for predictions of 2 scores, even when using the entire training dataset, accuracy shows significant fluctuations and maintains a lower level compared to LLMs.

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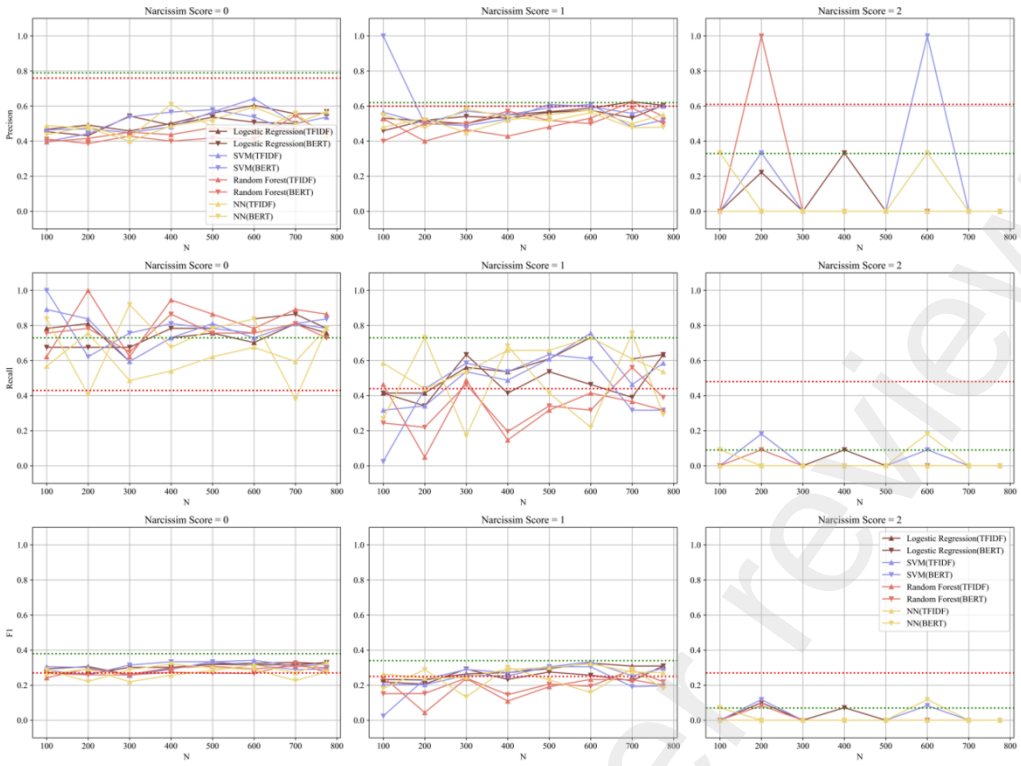


Figure 4 Answer -Level ML Model Metrics Change along with the sample of training data in CRT question (Green dash line is GPT040613 with few-shot, Red dash line is GPT040613 with zero-shot)

4.4 Comparison Analysis between LLMs Ratings and Expert Ratings

4.4.1 Sentence Confusion Analysis

In selecting highly-confusing response texts, we included all sentences where initial expert coding showed inconsistency, indicating these sentences were difficult for human experts to classify and prone to classification errors. In contrast, for low-confusion response texts, we selected texts from the test set where expert coding was consistent. We then compared ChatGPT's narcissism predictions between high-confusion and low-confusion texts. Table 6 presents the quantities of high-confusion and low-confusion sentences: in the SJT task, there are 179 low-confusion sentences and 166 high-confusion sentences; in the CRT task, there are 76 low-confusion sentences and 482 high-confusion sentences.

Table 6 Descriptive Statistical Table of Number of Sentences with Varying Levels of Confusion

	Training data		Testing data		consistent	inconsistent
	consistent	inconsistent	consistent	inconsistent		
SJT	0	130	179	36	179	166
CRT	0	438	76	44	76	482

The correct prediction rates of ChatGPT under conditions of high and low confusion were

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subjected to differential testing. Results for the SJT task are presented in Table 7, while those for the CRT task are shown in Table 8. The statistical test visualization is depicted in Figure 1. Across both SJT and CRT tasks, ChatGPT exhibited significantly higher correct classification rates for low-confusion texts compared to high-confusion texts. This suggests that like human experts, ChatGPT also struggles to classify texts correctly in categories where human experts are prone to errors.

Table 7 Statistical Test of Significant Differences in Correct Prediction Rates between High and Low Confusion Conditions - SJT

Models	Correct Prediction Rates		difference	Z	Cohens' d
	Low Confusion (N = 179)	High Confusion (N = 166)			
ChatGPT 3.5 – Zero-shot	114(64%)	58(35%)	0.29	5.34***	0.60
ChatGPT 3.5 – Few-shot	119(66%)	64(39%)	0.28	5.19***	0.58
ChatGPT 4 – Zero-shot	141(79%)	77(46%)	0.32	6.23***	0.71
ChatGPT 4 – Few-shot	156(87%)	67(40%)	0.47	9.08***	1.12

Table 8 Statistical Test of Significant Differences in Correct Prediction Rates between High and Low Confusion Conditions - CRT

Models	Correct Prediction Rates		difference	Z	Cohens' d
	Low Confusion (N = 76)	High Confusion (N = 482)			
ChatGPT 3.5 – Zero-shot	41(54%)	137(28%)	0.26	4.44***	0.56
ChatGPT 3.5 – Few-shot	46(61%)	137(28%)	0.32	5.54***	0.70
ChatGPT 4 – Zero-shot	53(70%)	145(30%)	0.40	6.71***	0.86
ChatGPT 4 – Few-shot	53(70%)	182(38%)	0.32	5.25***	0.66

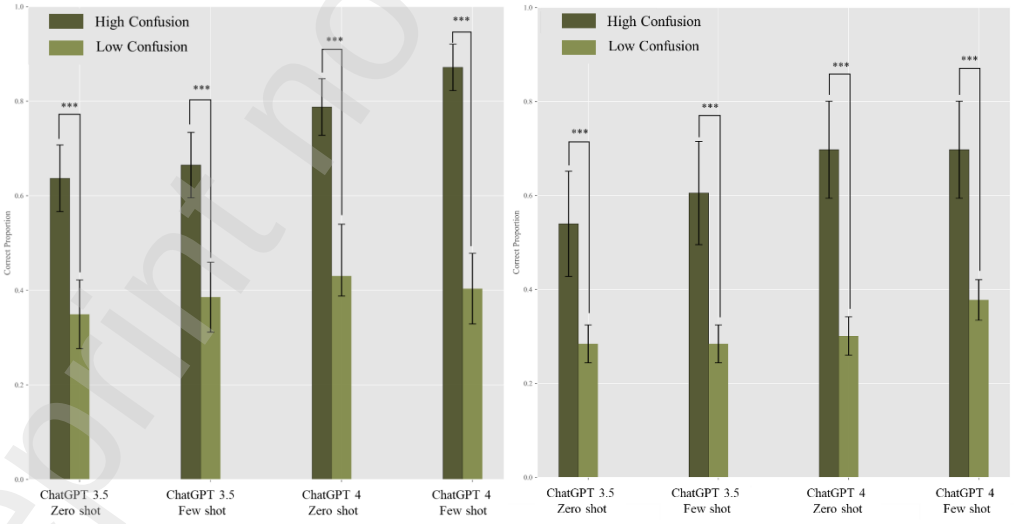


Figure 5 Plot of Proportion Differences of SJT (left) and CRT (right)

4.4.2 Error Pattern Analysis

In this section, we aim to analyze the types of errors made by human experts compared to

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ChatGPT, investigating whether error types common among human experts also frequently occur in ChatGPT's narcissism ratings. Table 9 and Table 10 respectively present the error classifications for ChatGPT and human experts in the SJT and CRT tasks. E1 denotes the misclassification of sentences belonging to a specific category into another category. E2 denotes the misclassification of sentences from other categories into a specific category

In the SJT task, human expert errors predominantly occur within three distinct sentence categories labeled as A1, A2, and A3. These categories are prone to both types of errors: misclassifying sentences belonging to these categories into other categories (E1 errors, constituting 72% of all E1 errors), and incorrectly classifying sentences from other categories into these specific categories (E2 errors, making up 75% of all E2 errors). ChatGPT exhibits a pattern of errors highly consistent with human experts. In E1 errors, ChatGPT also frequently misclassifies sentences within categories A1, A2, and A3. Across four different ChatGPT models, the cumulative proportion of misclassifications in these three categories ranges from 75% to 77%, while errors in other categories are comparatively lower.

In E2 errors, there is a noticeable difference between the ChatGPT 3.5 model and human experts. Apart from categories A1, A2, and A3, ChatGPT 3.5 also tends to misclassify sentences from other categories into category A10 more frequently compared to human experts. On the other hand, the ChatGPT 4 model aligns closely with human error patterns. The cumulative proportion of misclassification errors in categories A1, A2, and A3 for ChatGPT 4 is consistently around 71%. However, in addition to these categories, the ChatGPT 4 model also shows a tendency to misclassify sentences from other categories into A5, with proportions of 15% and 11%, respectively, which is higher than the 8% error rate observed in human experts.

Overall, in the SJT rating task, ChatGPT exhibits error types that are consistent with those made by human experts. Where human experts are prone to misclassification errors, ChatGPT similarly demonstrates susceptibility to these errors

Table 9 Comparison Analysis Table of Error Types in the SJT Task

Label	ChatGPT 3.5 （zero-shot）		ChatGPT 3.5 （few-shot）		ChatGPT 4 （zero-shot）		ChatGPT 4 （few-shot）		Human	
	E1	E2	E1	E2	E1	E2	E1	E2	E1	E2
A1	0.11	0.43	0.18	0.29	0.19	0.38	0.19	0.38	0.25	0.23

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A2	0.44	0.08	0.34	0.24	0.36	0.19	0.37	0.20	0.26	0.33
A3	0.20	0.09	0.22	0.09	0.22	0.14	0.19	0.13	0.21	0.19
A4	0.01	0.01	0.01	0.01	0.01	0.00	0.01	0.00	0.03	0.02
A5	0.03	0.08	0.04	0.06	0.01	0.14	0.05	0.11	0.08	0.08
A6	0.03	0.02	0.04	0.02	0.03	0.04	0.04	0.04	0.01	0.03
A7	0.03	0.00	0.04	0.01	0.04	0.00	0.02	0.01	0.03	0.02
A8	0.00	0.02	0.00	0.04	0.00	0.03	0.00	0.02	0.04	0.00
A9	0.07	0.08	0.07	0.07	0.07	0.03	0.07	0.06	0.08	0.05
A10	0.06	0.18	0.06	0.16	0.07	0.07	0.06	0.05	0.02	0.04

In the CRT task, human experts show a relatively even distribution of error types across all 12 categories, with E1 and E2 error proportions fluctuating around 10% for each category.

For ChatGPT models:

1) In E1 errors, there are variations among different ChatGPT models. ChatGPT 3.5 predominantly makes E1 errors in categories B1, B3, B4, and B5, accounting for a total error proportion of 83%. The ChatGPT 3.5 model with Few-shot learning shows a more evenly distributed pattern of E1 errors.
2) ChatGPT 4 and the ChatGPT model with Few-shot learning also exhibit relatively evenly distributed E1 error types, with a stronger concentration in categories B1 and B5, totaling error proportions of 45% and 41%, respectively.
3) In E2 error types, all four models demonstrate similarity to human experts, with error classification proportions evenly distributed across all categories.

Overall, in the CRT task, there is inconsistency in E1 errors between ChatGPT and human experts, but consistency in E2 errors. This indicates that while ChatGPT models may differ in the types of misclassifications compared to human experts, they generally align in terms of the overall distribution of errors in E2 classification types.

Table 10 Comparison Analysis Table of Error Types in the CRT Task

Label	ChatGPT 3.5 （zero-shot）		ChatGPT 3.5 （few-shot）		ChatGPT 4 （zero-shot）		ChatGPT 4 （few-shot）		Human	
	E1	E2	E1	E2	E1	E2	E1	E2	E1	E2
B1	0.20	0.03	0.08	0.05	0.22	0.04	0.20	0.04	0.09	0.03
B2	0.03	0.12	0.04	0.11	0.05	0.11	0.06	0.11	0.12	0.10

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B3	0.23	0.06	0.10	0.07	0.05	0.09	0.06	0.09	0.07	0.09
B4	0.20	0.11	0.10	0.16	0.13	0.12	0.13	0.14	0.15	0.15
B5	0.20	0.06	0.21	0.07	0.23	0.06	0.21	0.06	0.08	0.07
B6	0.06	0.09	0.15	0.06	0.11	0.07	0.10	0.08	0.11	0.12
B7	0.01	0.13	0.04	0.12	0.01	0.13	0.03	0.13	0.06	0.10
B8	0.02	0.07	0.00	0.09	0.03	0.07	0.02	0.08	0.06	0.06
B9	0.01	0.12	0.02	0.13	0.03	0.12	0.04	0.08	0.05	0.09
B10	0.03	0.13	0.06	0.10	0.04	0.13	0.03	0.11	0.09	0.12
B11	0.01	0.04	0.17	0.02	0.01	0.03	0.04	0.03	0.04	0.03
B12	0.01	0.04	0.04	0.03	0.08	0.03	0.08	0.03	0.07	0.03

5. Discussion

5.1 LLMs are effective tools for open-ended scoring of SJT and CRT texts.

In our study, we utilized ChatGPT 3.5 and ChatGPT 4, two prominent LLMs, to evaluate open-ended SJT and CRT responses. We then compared these scores with those generated by traditional machine learning models. The results support our research hypothesis, demonstrating that LLMs exhibit high accuracy in narcissism recognition under both Zero-shot and Few-shot conditions. While Zero-shot Learning performs slightly lower than Few-shot Learning, it still achieves performance comparable to top-performing machine learning models in specific tasks. This finding aligns with existing research on LLMs (Peters & Matz, 2023; Rathje et al., 2023; Yang et al., 2023).

Furthermore, our study reveals that LLMs outperform machine learning models in terms of specific score levels. LLMs demonstrate higher Precision and Recall in determining narcissism levels at scores 0, 1, and 2. In contrast, machine learning models exhibit lower Recall specifically in texts categorized with a score of 2. We attribute these results to the inherent advantages of LLMs. Machine learning models heavily rely on large-scale datasets (Pulicharla, 2024), making them less effective in small-sample datasets such as test texts. Additionally, scoring texts with a score of 2 on the narcissism scale is relatively rare in the population, posing a challenge for machine learning models to learn and correctly classify such texts. This aspect can also be observed from the dramatic fluctuations in the predictions of 2-point text(see Figure 1 – Figure 4). LLMs, however, are not constrained by dataset size limitations; they primarily leverage their existing knowledge (or understanding of narcissistic personality) and well-designed prompts to perform narcissism scoring tasks effectively.

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However, compared to SJT, the consistency of narcissism text scores generated under CRT is lower. Examining the content of responses, SJT responses typically fall into the "behavioral" category, where respondents describe how they would understand and address situational problems (Weekly & Ployhart, 2006; Rockstuhl et al., 2015). On the other hand, CRT responses belong to the "cognitive" category, where respondents state the basis, they believe best refutes a given argument (James, 1998). In terms of scoring processes, SJT tasks require raters to classify respondents' behaviors and anchor them into specific behavioral categories. In contrast, CRT tasks demand more from raters; they must infer implicit cognitive biases based on the arguments put forward by respondents (Galić et al., 2021; LeBreton et al., 2020; Schoen et al., 2021) and look beyond behavior to understand the essence. The differences in scoring difficulty also reflect in the variability of human expert scores. LLMs may perform less effectively in CRT tasks possibly due to their ability to infer implicit defense mechanisms based on thoughts, similar to theory of mind (Wester et al., 2024). However, more research is needed to confirm the relationship between CRT scoring and theory of mind.

5.2 LLMs possess rater-like characteristics.

Since the emergence of LLMs, many researchers have evaluated their human-like qualities from various perspectives, assessing whether they approach true human likeness. Current research has yielded diverse conclusions: some researchers evaluate LLMs' personalities, determining if they exhibit traits akin to human personalities, such as the Big Five personality traits (Karra et al., 2022), while others, like Huang (2023), conduct comprehensive evaluations of LLMs' personality, interpersonal relationships, motivation, and emotional capabilities. Some studies employ LLMs as subjects in psychological experiments to explore if they exhibit human-like behaviors under experimental conditions to assess their degree of human-like behavior. For instance, Binz & Schulz (2023) used LLMs in economic decision-making tasks, and Aher et al. (2023) replicated Milgram's obedience experiments using LLMs.

The degree of human-likeness of LLMs is a topic of widespread interest among researchers. However, conclusive assessments remain elusive, necessitating further research. Perhaps more refined tasks are needed to determine where LLMs exhibit human-like capabilities and where they do not. In our evaluation of narcissism text scoring, LLMs have demonstrated a high degree of

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human-likeness compared to human expert ratings. Comparing LLMs' scoring patterns with those of human experts, we find two main similarities: firstly, both LLMs and human experts struggle to accurately classify texts that are difficult for human experts; secondly, in cases of misclassification, LLMs tend to make errors similar to those made by human experts, indicating a consistent pattern of misclassification between LLMs and human experts.

5.3 Designing a narcissistic personality assessment tool based on LLMs scoring.

Accurate assessment of narcissism is crucial in clinical and organizational management settings. However, due to its negative connotations, self-report measures of narcissism are often vulnerable to influences such as self-deception and social desirability (Sackeïm, 1979; van de Mortel, 2008). This vulnerability is particularly pronounced in high-stakes selection scenarios where respondents may have strong motivations to present themselves in a more favorable light to gain competitive advantages (Chan, 2010). Assessments conducted in the form of open-ended questions can mitigate these biases. However, relying on trained psychologists to individually evaluate each response incurs high costs.

Our research has demonstrated the potential of LLMs in identifying narcissistic traits, particularly in open-ended SJT tasks. By training LLMs to function as specialized experts in narcissistic personality recognition, we can address the cost issue associated with scoring open-ended questions. Through well-designed prompts and examples that typify narcissistic behaviors, LLMs can achieve scoring capabilities comparable to human experts. Our study has preliminarily validated the feasibility of applying LLMs to scoring open-ended narcissism texts: it is cost-effective, accurate in scoring, and shows generalizability across multiple tasks.

Future efforts should focus on developing comprehensive tools for open-ended narcissistic personality assessment integrated with LLMs scoring, validating their reliability and validity, and examining predictive validity against actual behavioral outcomes.

5.4 Limitations

Our study aimed to explore the capability of LLMs in identifying narcissistic traits in SJT and CRT test texts, utilizing specifically the OpenAI released ChatGPT 3.5 and ChatGPT 4 models. It is important to note that our findings may not fully represent the general ability level of LLMs. Additionally, in constructing comparative machine learning models, we only utilized TF-IDF values

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and BERT word embeddings for text features, without considering more complex machine learning model construction techniques. It is possible that more sophisticated machine learning models could perform better in these tasks. Furthermore, in reviewing existing literature on CRT, we are the first researchers to adapt CRT tests into open-ended responses. The inconsistency in scoring within CRT tasks may be attributed to the test itself not being suitable for adaptation into open-ended responses.

6. Conclusion

Our study investigated the capacity of two large language models (LLMs), specifically ChatGPT 3.5 and ChatGPT 4, to identity narcissistic traits in open-ended situational judgment tests (SJT) and conditional reasoning tests (CRT). The findings demonstrate that under zero-shot learning conditions, LLMs perform comparably to top-performance machine learning models. Few-shot learning effectively enhances the accuracy of LLMs in identifying narcissistic traits in Sentence-Level Model but reduce the accuracy in Answer-Level Model, with SJT texts being evaluated more accurately than CRT texts. Through detailed analysis, we observed that LLMs exhibit scoring performances highly congruent with human experts: they make judgments identical to experts on texts where there is unanimous expert agreement, and they also tend to make errors on texts where experts disagree. Further analysis of error patterns confirms that LLMs exhibit error patterns similar to human experts. In conclusion, our study establishes that LLMs can accurately identify narcissistic traits in open-ended SJT and CRT test texts, contributing to the growing body of research on artificial intelligence in psychological assessment.

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Appendix

TableA1 Sentences labels for open-ended situation judgement test

Label	Description	Example sentences	Counts
Analyze the reasons for failure	hold a future orientation that sees failure as the mother of success, explore reasons for failure	Take it seriously, identify my own shortcomings. Find the reasons for management failure	660
Accepting failure calmly	Accept failure calmly, admit himself' shortcomings	It's a matter of lacking skills compared to others, which is an issue of my own abilities.	161
Adaptive emotional adjustment	Use some means to relieve negative emotions, which include exercise, walking, games, self-encouragement, etc.	I will alleviate this emotion by shifting my focus, such as engaging in outdoor activities, listening to music, and watching movies	214
Seeking social support	Try to seek comfort from friends or family.	I will tell this to my friends, to relieve my own sense of loss and gain encouragement from them.	63
Compensation	Demonstrate his abilities to colleagues in other ways, which include cite previous successful experiences to prove himself/herself or prove himself in future job	I will use my own way, stick to my beliefs, persevere, until I succeed. I will use actions to prove that my ideas are valid.	90
Dismiss	Ignore colleagues' criticism, don't care what colleagues think	There is no need to pay attention to the discussions and doubts of colleagues within the company	40
Avoidance	Avoid the stimuli associated with the setback event and reduce communication with colleagues, hope to escape the scene of failure (such as quitting)	I dare not chat with my colleagues as usual; there is a sense of avoidance.	50
Shame blaming	Relieve shame emotions by blaming or belittling others who are not as good as oneself	"If you can do it, then go ahead." After all, they don't even have the qualifications to give it a try.	14
Emotion states	Only state emotions he/she feels	Feeling lost and having doubts about my work abilities.	632
Invalid response	Meaningless phrases or repetitive characters		42

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TableA2 Sentences labels for open-ended conditional reasoning test

Label	Example sentences	Counts
Exceptional talent won't make people feel inferior	The way elites showcase their talents is modest and gentle, not making ordinary people feel uncomfortable or inferior	41
The elites are excluded because of a lack of interpersonal skills	A person with exceptional talent may be excluded due to a lack of social skills	157
The elite and the ordinary are no different	No one is born an elite; elites are created through step-by-step cultivation from ordinary individuals	136
Ordinary people tend to learn from the elite	individuals tend to admire and learn from elites rather than envy and ostracize them	185
Elites can harmoniously coexist with ordinary people	Many elites also having better affinity, being approachable, and able to relate well to ordinary people	52
The elite attract admiration and adoration from ordinary people	People often follow and chase after elites as their role models	145
The exclusion of elites is due to the narrow-mindedness of ordinary people.	Excellent people are inevitably envied by ordinary people	107
Elites rarely have opportunities to interact with ordinary people.	The daily life of elites mainly interacting with fellow elites	77
Elites are generally of high moral character and will not be excluded	The elite are not just exceptionally talented. I believe they have their own standards for how they conduct themselves. At the very least, they can be polite and humble towards everyone	73
The elite can make significant contributions to society and should not be excluded	The wisdom of elites should be a cherished resource in the world, not subject to unfair exclusion	116
The ordinary person will benefit from their relationship with the elite	Following elites can allow oneself to gain greater benefits	25
Invalid response		29

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